

DGFF Lecture 1 NOTE

0. Mental model / my personal summary

Core philosophy

DGFF is *not* first defined by the law of each individual h_x . It is first a *joint law* on the whole configuration

$$h = (h_x)_{x \in V} \in \mathbb{R}^V.$$

Then each h_x is just a coordinate projection.

Three equivalent faces of the same object:

$$\boxed{\text{energy / Gibbs law}} \iff \boxed{\text{Gaussian with covariance } G^V} \iff \boxed{\text{Hilbert-space Gaussian in Dirichlet space}}.$$

The same quadratic form reappears everywhere:

$$\sum_{x \sim y} (h_x - h_y)^2.$$

It is simultaneously:

- the **energy** in the Gibbs density,
- the **Dirichlet energy** in electrical networks,
- the quadratic form of the **Dirichlet Laplacian**,
- the source of the **Green function** after inversion.

1. Setup

Let $V \subset \mathbb{Z}^d$ be finite unless otherwise stated.

Boundary convention: $h_x = 0$ for $x \notin V$ (zero boundary condition).

Simple random walk:

$$X_0 = x, \quad \mathbb{P}(X_{n+1} = y \mid X_n = z) = \frac{1}{2d} \mathbf{1}_{\{y \sim z\}}.$$

Exit time:

$$\tau_{V^c} := \inf\{n \geq 0 : X_n \notin V\}.$$

Green function in V :

$$G^V(x, y) := \mathbb{E}^x \left[\sum_{n=0}^{\tau_{V^c}-1} \mathbf{1}_{\{X_n=y\}} \right].$$

Interpretation: $G^V(x, y)$ is the expected number of visits to y before exiting V , starting from x .

2. Construction I: explicit Gibbs / energy definition

$$\mathbb{P}(h \in A) = \frac{1}{Z_V} \int_A \exp\left(-\frac{1}{4d} \sum_{(x,y) \in E(\mathbb{Z}^d)} (h_x - h_y)^2\right) \prod_{x \in V} dh_x \prod_{x \notin V} \delta_0(dh_x).$$

Key understanding:

- This defines a density on the *whole configuration* h , not on each h_x separately.
- $\prod_{x \notin V} \delta_0(dh_x)$ means all coordinates outside V are fixed to 0.
- Therefore the true degrees of freedom live in $\mathbb{R}^V$.
- The weight is larger for configurations with small edge differences.

Energy / Hamiltonian:

$$\mathcal{E}(h) := \frac{1}{4d} \sum_{x \sim y} (h_x - h_y)^2.$$

Hence

$$\mathbb{P}(h) \propto e^{-\mathcal{E}(h)}.$$

Why boundary condition is necessary

Without fixing boundary, the energy is invariant under global shifts:

$$h_x \mapsto h_x + c \implies (h_x + c) - (h_y + c) = h_x - h_y.$$

So the density would not be integrable along the constant direction. Boundary condition kills this zero mode.

3. Construction II: via the Green function

Define DGFF in V as the centered Gaussian field

$$h^V = \{h_x^V : x \in \mathbb{Z}^d\}$$

such that

$$\mathbb{E}(h_x^V) = 0, \quad \mathbb{E}(h_x^V h_y^V) = G^V(x, y).$$

Equivalently,

$$h^V \sim \mathcal{N}(0, G^V).$$

Key understanding:

- This is a Gaussian *joint law*, not a separate definition of each h_x .
- Individual marginals are consequences:

$$h_x^V \sim N(0, G^V(x, x)).$$

4. Construction III: Hilbert-space Gaussian

Define

$$\mathcal{H}^V := \{f : \mathbb{Z}^d \rightarrow \mathbb{R}, \text{ supp}(f) \subset V\}.$$

Dirichlet inner product:

$$\langle f, g \rangle_{\nabla} := \frac{1}{2d} \sum_{x \in \mathbb{Z}^d} \nabla f(x) \cdot \nabla g(x),$$

where

$$\nabla f(x) = (f(x + e_1) - f(x), \dots, f(x + e_d) - f(x)).$$

Then $\mathcal{H}^V$ is a finite-dimensional Hilbert space.

Let $\{\phi_n\}_{n=1}^{|V|}$ be an orthonormal basis in $(\mathcal{H}^V, \langle \cdot, \cdot \rangle_{\nabla})$, and let $Z_1, \dots, Z_{|V|}$ be i.i.d. $N(0, 1)$. Define

$$\tilde{h}_x := \sum_{n=1}^{|V|} \phi_n(x) Z_n.$$

Then $\tilde{h} = \{\tilde{h}_x\}_{x \in \mathbb{Z}^d}$ has the law of the DGFF in V .

Why this is Gaussian

For any finite set $x_1, \dots, x_k$,

$$(\tilde{h}_{x_1}, \dots, \tilde{h}_{x_k}) = A(Z_1, \dots, Z_{|V|})^T$$

for a deterministic matrix A , so it is multivariate Gaussian.

Covariance computation

$$\mathbb{E}(\tilde{h}_x \tilde{h}_y) = \sum_{n=1}^{|V|} \phi_n(x) \phi_n(y).$$

Then one proves

$$\sum_{n=1}^{|V|} \phi_n(x) \phi_n(y) = G^V(x, y),$$

so indeed

$$\tilde{h} \sim \mathcal{N}(0, G^V).$$

5. The Laplacian and the inverse-kernel viewpoint

Discrete Laplacian

$$\Delta f(x) := \sum_{y \sim x} (f(y) - f(x)).$$

Thus

$$(-\Delta f)(x) = \sum_{y \sim x} (f(x) - f(y)).$$

With zero boundary condition on V^c , define the Dirichlet Laplacian

$$L := -\Delta_D.$$

Green function solves a Poisson problem

For fixed x ,

$$\begin{cases} \Delta G^V(x, \cdot) = -2d \delta_x(\cdot), & \text{on } V, \\ G^V(x, \cdot) = 0, & \text{on } V^c. \end{cases}$$

Equivalently,

$$-\Delta G^V(x, \cdot) = 2d \delta_x.$$

Hence

$$G^V = 2d(-\Delta_D)^{-1}.$$

Meaning of “functional-analytic representation”

The probabilistic object

$$G^V(x, y) = \mathbb{E}^x \left[\sum_{n=0}^{\tau_{V^c}-1} \mathbf{1}_{\{X_n=y\}} \right]$$

can also be viewed as the kernel of the inverse Dirichlet Laplacian:

$$G^V(x, y) = 2d(-\Delta_D)^{-1}(x, y).$$

So Green function has three equivalent meanings:

1. expected occupation time,
2. inverse Laplacian kernel,
3. grounded point-source potential in electrical networks.

6. The master quadratic-form identity

For $h \in \mathbb{R}^V$ (extended by 0 on V^c),

$$h^T L h = \sum_{x \sim y} (h_x - h_y)^2.$$

Proof idea: Expand

$$h^T L h = \sum_{x \in V} h_x \sum_{y \sim x} (h_x - h_y).$$

For an interior edge $\{x, y\}$ the two contributions add up:

$$h_x(h_x - h_y) + h_y(h_y - h_x) = (h_x - h_y)^2.$$

For a boundary edge with $y \notin V$,

$$h_x(h_x - h_y) = h_x^2 = (h_x - 0)^2.$$

Hence the Gibbs density becomes

$$\exp\left(-\frac{1}{4d} h^T L h\right) = \exp\left(-\frac{1}{2} h^T \left(\frac{1}{2d} L\right) h\right).$$

Therefore the precision matrix is

$$Q = \frac{1}{2d} L,$$

and the covariance matrix is

$$Q^{-1} = 2d L^{-1} = G^V.$$

This proves Construction I = Construction II.

7. Why the energy is so important

$$\mathcal{E}(f) = \frac{1}{4d} \sum_{x \sim y} (f(x) - f(y))^2.$$

This same quantity appears in three ways:

1. Probability:

$$\mathbb{P}(h) \propto e^{-\mathcal{E}(h)}.$$

DGFF samples configurations according to roughness.

2. Analysis:

$$\langle f, (-\Delta_D) f \rangle = \sum_{x \sim y} (f(x) - f(y))^2.$$

Energy is the quadratic form of the Dirichlet Laplacian.

3. Electrical networks: edge differences are voltage drops, and energy is dissipated energy.

My personal summary:

$$\boxed{\text{energy}} \rightarrow \boxed{\text{Laplacian}} \rightarrow \boxed{\text{inverse}} \rightarrow \boxed{\text{Green function}} \rightarrow \boxed{\text{DGFF covariance}}.$$

8. Electrical network interpretation

For grounded boundary V^c :

- If one injects **unit current** at x , then the resulting voltage profile is

$$u_x(\cdot) = \frac{1}{2d} G^V(x, \cdot).$$

- Therefore

$$G^V(x, x) = 2d R_{\text{eff}}(x \leftrightarrow V^c).$$

If instead one normalizes by **voltage**:

$$v_x(y) := \mathbb{P}^y(\tau_x < \tau_{V^c}),$$

then

$$v_x(x) = 1, \quad v_x = 0 \text{ on } V^c,$$

and

$$G^V(x, y) = G^V(x, x) v_x(y).$$

Important distinction:

- unit *current* normalization gives Green function naturally;
- unit *voltage* normalization gives hitting probability / equilibrium potential.

9. Key board-proof facts

(A) Exercise 1.2: integrability of the Gibbs density

Need to show

$$\int_{\mathbb{R}^V} \exp\left(-\frac{1}{4d} \sum_{x \sim y} (h_x - h_y)^2\right) dh < \infty.$$

Idea:

- The quadratic form

$$Q(h) := \sum_{x \sim y} (h_x - h_y)^2$$

is positive definite on $\mathbb{R}^V$ because boundary values are pinned to 0.

- Hence

$$Q(h) \geq c \sum_{x \in V} h_x^2$$

for some $c > 0$.

- Therefore density is bounded by an integrable Gaussian:

$$e^{-Q(h)/(4d)} \leq e^{-c|h|^2/(4d)}.$$

(B) Exercise 1.4: Green solves Poisson

Fix x and let $g(y) = G^V(x, y)$.

For $y \neq x$,

$$g(y) = \frac{1}{2d} \sum_{z \sim y} g(z),$$

so

$$\Delta g(y) = 0.$$

For $y = x$,

$$g(x) = 1 + \frac{1}{2d} \sum_{z \sim x} g(z),$$

hence

$$\Delta g(x) = -2d.$$

Therefore

$$\Delta G^V(x, \cdot) = -2d \delta_x \quad \text{on } V.$$

(C) Exercise 1.5(1): symmetry

Since

$$G^V = 2d (-\Delta_D)^{-1},$$

and $-\Delta_D$ is symmetric, so is its inverse:

$$G^V(x, y) = G^V(y, x).$$

(D) Exercise 1.5(2): positivity

For finite-support f ,

$$\sum_{x, y} G^V(x, y) f(x) f(y) = 2d \langle f, (-\Delta_D)^{-1} f \rangle_{\ell^2} \geq 0.$$

More concretely, if $u = (-\Delta_D)^{-1} f$, then

$$\sum_{x, y} G^V(x, y) f(x) f(y) = 2d \sum_x u(x) (-\Delta_D u)(x) = d \sum_{x \sim y} (u(x) - u(y))^2 \geq 0.$$

(E) Exercise 1.7: equivalence of Def. 1.1 and Def. 1.6

By the quadratic-form identity,

$$\mathbb{P}(h) \propto \exp\left(-\frac{1}{4d}h^T L h\right) = \exp\left(-\frac{1}{2}h^T \left(\frac{1}{2d}L\right)h\right).$$

So the law is centered Gaussian with precision matrix $\frac{1}{2d}L$, hence covariance

$$\left(\frac{1}{2d}L\right)^{-1} = 2dL^{-1} = G^V.$$

Thus it is exactly $\mathcal{N}(0, G^V)$.

10. Why the Dirichlet inner product is really an inner product

$$\langle f, g \rangle_{\nabla} = \frac{1}{2d} \sum_x \nabla f(x) \cdot \nabla g(x).$$

- Bilinear: obvious from linearity of ∇ .
- Symmetric: Euclidean dot product is symmetric.
- Positive:

$$\langle f, f \rangle_{\nabla} = \frac{1}{2d} \sum_x |\nabla f(x)|^2 \geq 0.$$

- Definite: if $\langle f, f \rangle_{\nabla} = 0$, then $\nabla f(x) = 0$ for all x , so f is constant along every lattice edge, hence constant on $\mathbb{Z}^d$. Since f has finite support (indeed $f = 0$ on V^c), that constant must be 0.

11. Why the Hilbert-space covariance is Green

For each fixed $x \in V$ and all $f \in \mathcal{H}^V$,

$$\langle G^V(x, \cdot), f \rangle_{\nabla} = f(x).$$

Reason:

$$\langle G^V(x, \cdot), f \rangle_{\nabla} = \frac{1}{2d} \sum_{z \in V} f(z) (-\Delta G^V(x, \cdot))(z) = \frac{1}{2d} \sum_{z \in V} f(z) 2d \delta_x(z) = f(x).$$

Now if $\{\phi_n\}$ is an orthonormal basis in $\mathcal{H}^V$, expand

$$G^V(x, \cdot) = \sum_{n=1}^{|V|} \langle \phi_n, G^V(x, \cdot) \rangle_{\nabla} \phi_n = \sum_{n=1}^{|V|} \phi_n(x) \phi_n(\cdot).$$

Hence

$$G^V(x, y) = \sum_{n=1}^{|V|} \phi_n(x) \phi_n(y).$$

Therefore for

$$\tilde{h}_x = \sum_{n=1}^{|V|} \phi_n(x) Z_n,$$

we get

$$\mathbb{E}(\tilde{h}_x \tilde{h}_y) = G^V(x, y),$$

so $\tilde{h}$ is exactly the DGFF.

12. Useful one-line oral-exam answers

Q: What is a random field? A family of random variables indexed by space:

$$\{X_x : x \in \mathbb{Z}^d\}.$$

Q: What is a Gaussian process/field? Every finite subfamily is jointly Gaussian.

Q: Why is DGFF a joint object, not a pointwise one? Because the law is defined on the whole configuration $h = (h_x)_{x \in V}$ through a coupled energy depending on edge differences.

Q: Why is Green function relevant? Because covariance of a Gaussian is the inverse of the precision operator, and the precision operator here is the Dirichlet Laplacian.

Q: Why does Green function solve a Poisson equation? Because occupation times satisfy first-step decomposition; away from the source the function is harmonic, and at the source there is an extra initial visit.

Q: Why is the Hilbert-space construction natural? Because the DGFF energy induces the Dirichlet inner product, so DGFF is the standard Gaussian associated with that Hilbert geometry.

13. My own conceptual chain

What I should remember under pressure

DGFF is a Gaussian measure on configurations

with three interchangeable descriptions:

$$\mathbb{P}(h) \propto \exp\left(-\frac{1}{4d} \sum_{x \sim y} (h_x - h_y)^2\right)$$

$$\mathbb{E}(h_x h_y) = G^V(x, y)$$

$$h = \sum_n Z_n \phi_n \quad \text{in the Dirichlet Hilbert space}$$

and all three are glued together by

$$G^V = 2d(-\Delta_D)^{-1}.$$

If I forget everything else, I should reconstruct from:

energy $\leftrightarrow$ Laplacian $\leftrightarrow$ inverse $\leftrightarrow$ Green $\leftrightarrow$ Gaussian covariance.

14. Dimensional picture, $d = 2$, harmonic measure, and my current understanding

14.1 What the second half of Lecture 1 is really doing

At first I thought the later part of Lecture 1 was just giving several side results: Brownian bridge in $d = 1$, thermodynamic limit in $d \geq 3$, and then some technical analysis in $d = 2$. Now I think the real question is:

As the domain size tends to infinity, what is the correct macroscopic incarnation of the DGFF?

The point is that the answer depends completely on dimension.

$d = 1$: pointwise variance is of order N , so there is no meaningful pointwise infinite-volume limit. The right object is a *scaling limit*: after diffusive rescaling one gets a Brownian bridge.

$d \geq 3$: pointwise variance stays bounded, so one can hope for a pointwise infinite-volume Gaussian field. This is the *thermodynamic limit*: the lattice stays fixed and the box grows.

$d = 2$: this is the critical case. Variance diverges, but only logarithmically:

$$G_{V_N}(x, x) \sim g \log N, \quad g = \frac{2}{\pi}.$$

So $d = 2$ is neither like $d = 1$ nor like $d \geq 3$: the pointwise thermodynamic limit fails, but the naive pointwise scaling limit also fails. This is exactly why the lecture becomes much more subtle here.

14.2 My current conceptual summary of the dimensional trichotomy

The later part of Lecture 1 is not a collection of unrelated examples. It is a classification of the possible large-volume behaviors of the DGFF:

$$\begin{aligned} d = 1 & : \text{ too rough pointwise } \Rightarrow \text{ rescale to a continuum path object,} \\ d = 2 & : \text{ critical } \Rightarrow \text{ no pointwise limit, need a distribution-valued limit,} \\ d \geq 3 & : \text{ stable pointwise } \Rightarrow \text{ infinite-volume lattice field.} \end{aligned}$$

So the role of the $d = 1$ and $d \geq 3$ discussions is really to isolate why $d = 2$ is special.

14.3 Brownian bridge in $d = 1$ and why it appears

For $V_N = (0, N) \cap \mathbb{Z}$, the one-dimensional DGFF has energy

$$\frac{1}{4} \sum_{x=0}^{N-1} (h_{x+1} - h_x)^2, \quad h_0 = h_N = 0.$$

If I define increments

$$\eta_x := h_{x+1} - h_x, \quad x = 0, \dots, N-1,$$

then the field can be viewed as a Gaussian random walk conditioned on

$$\sum_{x=0}^{N-1} \eta_x = 0.$$

So the DGFF in $d = 1$ is essentially a Gaussian walk pinned at both endpoints. That is why the scaling limit is not Brownian motion but Brownian bridge:

$$\left\{ \frac{1}{\sqrt{N}} h_{\lfloor tN \rfloor} : t \in [0, 1] \right\} \Longrightarrow \{ \sqrt{2} W_t : t \in [0, 1] \},$$

where W is the standard Brownian bridge.

Important conceptual point: the bridge structure is just the continuum manifestation of the boundary pinning $h_0 = h_N = 0$.

14.4 Tightness: my interpretation

A family (X_n) of real-valued random variables is tight if for every $\varepsilon > 0$, there exists $M < \infty$ such that

$$\sup_n P(|X_n| > M) < \varepsilon.$$

My own interpretation is that tightness is a kind of *probabilistic uniform control*. It feels analogous to the idea of a uniform bound in analysis: one wants a *single* bounded region that contains almost all of the mass for the *entire family*.

So I think of:

tightness as a probabilistic analogue of uniform boundedness / uniform control.

In particular, for centered Gaussian random variables, if the variances blow up, then the family is not tight. This explains the sentence in the lecture that in $d = 2$ there is no pointwise thermodynamic limit:

$$\text{Var}(h_x^{V_N}) = G^{V_N}(x, x) \rightarrow \infty,$$

so the point values cannot have a nontrivial tight limit.

14.5 Why the naive normalization by $\sqrt{\log N}$ fails in $d = 2$

A natural idea is to normalize by the square-root of the variance:

$$\tilde{h}_x^{V_N} := \frac{h_x^{V_N}}{\sqrt{\log N}}.$$

At first this looks reasonable because the pointwise variance is of order $\log N$.

But the key observation is that if two points are macroscopically separated, then

$$\sup_{N \geq 1} \sup_{\substack{x, y \in V_N \\ |x-y| \geq \varepsilon N}} G^{V_N}(x, y) < \infty.$$

So off-diagonal covariances do *not* grow like $\log N$. After dividing by $\sqrt{\log N}$, the normalized covariance between two distinct macroscopic points goes to 0.

Hence pointwise normalization destroys the spatial structure: it keeps the diagonal scale but washes out the correlations. So the problem is not just that the variance diverges, but that the divergence is concentrated on the diagonal.

My current summary: in $d = 2$, the field is too singular to be observed pointwise, but the non-diagonal structure is still meaningful.

14.6 The correct viewpoint in $d = 2$: field as a distribution

The lecture then switches to a new method: instead of looking at point values, one looks at averages against test functions. That is, one studies quantities of the form

$$h^{D_N}(f),$$

rather than $h^{D_N}([xN])$.

This is the key conceptual jump: the limiting object is not an ordinary random function $x \mapsto h(x)$, but a random distribution

$$f \mapsto h(f).$$

My own way to think about this: the singularity in $d = 2$ lives on the diagonal. Point evaluation hits that singularity directly. Averaging against a test function spreads things out and makes the logarithmic singularity integrable.

So the field survives in a weaker sense: not as a pointwise object, but as a generalized Gaussian field.

14.7 Continuous domains D and discrete approximations D_N

I now understand that:

- $D \subset \mathbb{C}$ is a bounded, open, connected planar domain with reasonably non-pathological boundary;
- $D_N \subset \mathbb{Z}^2$ is not supposed to look like D directly, but rather

$$\frac{1}{N} D_N$$

is supposed to approximate D ;

- so D_N should be thought of as a lattice approximation of the blown-up domain ND .

The admissibility conditions say roughly:

- D_N does not stick outside D when rescaled;
- D_N eventually contains every part of D that stays a fixed positive distance away from the boundary.

So the approximation is an inside/outside control near the boundary, not just a naive discretization.

14.8 Harmonic measure: my interpretation

For Brownian motion (B_t) started from $x \in D$, define

$$\tau_{D^c} := \inf\{t \geq 0 : B_t \notin D\},$$

and

$$\Pi^D(x, A) := P^x(B_{\tau_{D^c}} \in A), \quad A \subset \partial D.$$

This is the harmonic measure from x in D .

My interpretation is:

$$\Pi^D(x, \cdot) = \text{the distribution of the exit point of Brownian motion from } D \text{ started at } x.$$

If f is boundary data, then

$$u(x) := \int_{\partial D} f(z) \Pi^D(x, dz) = E^x[f(B_{\tau_{D^c}})]$$

produces a harmonic function in D .

My analogy with the discrete case: this is the continuum analogue of the random-walk construction of voltage functions on a graph. In the discrete setting, one builds a harmonic function by taking the expected boundary value at the exit point of a random walk. In the continuum setting, Brownian motion plays the role of random walk, and harmonic measure is the exit distribution that carries the boundary data back into the interior.

So to me,

$$\text{random walk exit law} \leftrightarrow \text{Brownian harmonic measure}$$

is the discrete/continuous version of the same mechanism.

14.9 Potential kernel and the Green function representation

A key lemma in the lecture states that for finite $V \subset \mathbb{Z}^2$,

$$G^V(x, y) = -a(x - y) + \sum_{z \in \partial V} H^V(x, z) a(z - y),$$

where $H^V(x, z)$ is the exit distribution of the random walk from V .

My understanding is:

- $-a(x - y)$ is the free-space singular part;
- the boundary average term is the harmonic correction needed to enforce the boundary condition.

This is exactly parallel to the continuum picture:

$$\text{singular kernel} + \text{harmonic boundary correction}.$$

Important warning from Exercise 1.20: this finite-domain formula does *not* automatically extend to infinite V . For

$$V = \mathbb{Z}^2 \setminus \{0\},$$

one gets

$$G^{\mathbb{Z}^2 \setminus \{0\}}(x, y) = a(x) + a(y) - a(x - y),$$

so there is an extra term $a(x)$. This is exactly a sign that “boundary at infinity” is subtle and cannot be ignored.

14.10 My earlier difficulty with Ofer’s old problem, now reinterpreted

This connects directly to the old problem I discussed with Ofer: trying to compute the effective resistance between $(0, 0)$ and $(0, 1)$ in the full lattice $\mathbb{Z}^2$.

Originally, the tempting idea was to superpose two infinite network configurations. But the rigorous problem was:

In an infinite network, one cannot simply assume that the relevant potentials/currents exist in the needed class, are unique, and can be freely superposed.

So my real difficulty was never the numerical value itself. The real difficulty was the legitimacy of doing linear superposition in an infinite network without first building the right potential-theoretic framework.

Now, after reading this lecture, I think the organizing object was really the potential kernel / Green function all along. What looked like a simple resistance problem was already forcing me to confront:

- boundary conditions at infinity,
- infinite-volume potential theory,
- and the logarithmic structure special to $d = 2$.

This is one of the most important personal takeaways from my recent study.

14.11 Why $a(x) \sim \frac{2}{\pi} \log |x|$ is the heart of the $d = 2$ theory

The lecture then invokes the asymptotic

$$a(x) = g \log |x| + c_0 + O(|x|^{-2}), \quad g = \frac{2}{\pi}.$$

At this point I do not yet have a fully clean proof of the full statement, but I understand the meaning and heuristic:

- the potential kernel is the two-dimensional replacement for the full-space Green function, which diverges;
- in dimension 2, the basic singularity of the inverse Laplacian is logarithmic;
- therefore the potential kernel must grow like $\log |x|$.

This is the source of all the logarithms that appear later:

$$\text{potential kernel} \Rightarrow \text{Green asymptotic} \Rightarrow \text{DGFF variance} \Rightarrow \text{CGFF logarithmic covariance}.$$

My current summary: the real technical heart of the later part of Lecture 1 is not merely proving identities, but identifying the *scale* of the two-dimensional potential theory. Once one knows that the basic object grows like $\log$, all the later structures begin to make sense.

14.12 Martingales and harmonicity: the proof pattern I now recognize

A very important proof pattern in this lecture is:

1. construct a function ϕ that is discrete harmonic in the domain;
2. observe that

$$\phi(X_{\tau_{V^c} \wedge n})$$

is a martingale along the random walk;

3. apply Optional Stopping to get a boundary representation formula.

I now think of this as the discrete analogue of the continuum harmonic-measure representation. In both cases, the same principle is at work:

$$\text{harmonicity} \iff \text{mean-value property} \iff \text{martingale / exit-distribution representation}.$$

This is one of the clearest places where probability, potential theory, and electrical network intuition all become the same thing.

14.13 Personal analogies that I want to remember

These are my own conceptual analogies that have helped me organize the lecture:

- **Tightness as probabilistic uniform control.** It feels analogous to uniform boundedness/equicontinuity in analysis: one wants one uniform control parameter for the whole family, but only in probability.
- **Harmonic measure as the continuum exit law.** It is the continuous version of the boundary hitting distribution of a random walk.
- **Voltage construction via random motion.** In the discrete setting, random walk exit distributions build voltage functions. In the continuum setting, Brownian motion exit distributions build harmonic functions.
- **The field in $d = 2$ as too singular pointwise, but still meaningful after averaging.** Point evaluation is too sharp; test-function averaging smooths the diagonal singularity.
- **Potential kernel as the hidden organizing object behind older resistance problems.** What I originally saw as a network calculation is actually a question in two-dimensional potential theory.

14.14 Current difficulties / where I need help

At this point I think I understand the conceptual architecture of the second half of Lecture 1 quite well, but the most technical steps are still difficult for me to prove cleanly:

- the exact proof of the potential-kernel asymptotic

$$a(x) = \frac{2}{\pi} \log |x| + c_0 + O(|x|^{-2});$$

- the most technical parts of the Green function asymptotic in general domains;
- the precise transition from the discrete Green representation to the continuum kernel involving harmonic measure.

So my situation is:

- conceptually I now see why these steps matter;
- technically I still need guidance in carrying out the harder asymptotic estimates.

14.15 What I now think is the main message of Lecture 1

If I had to summarize the whole lecture in one sentence now, I would say:

The DGFF is a Gaussian object whose true structure is controlled by Green functions, and in dimension 2 the logarithmic asymptotics of the Green/potential kernel force the field to be understood not pointwise but as a distribution-valued continuum object.

Equivalently, the discrete and continuum stories are glued together by

$$\text{random walk / Brownian motion} \leftrightarrow \text{harmonicity} \leftrightarrow \text{Green function} \leftrightarrow \text{Gaussian field}.$$

14.16 My present mental reconstruction chain

If I forget details under pressure, this is the chain I want to reconstruct from:

$$\text{DGFF covariance} = \text{Green function}$$

$$\text{Green function} = \text{singular part} + \text{harmonic boundary correction}$$

$$\text{in } d = 2, \quad \text{singular part} \sim -\frac{2}{\pi} \log |x - y|$$

therefore pointwise values diverge logarithmically,

so the continuum limit must be distribution-valued rather than pointwise.

This is, at the moment, my best high-level summary of where the lecture is going.